Unit 1

Lesson Three

# Derivatives of exponential and logarithmic functions

## Derivative of exponential function

**1.** If $f(x) = e^x$ , then $\grave{f}(x) = e^x$

**2.** If $f(x) = a^x$ , then $\grave{f}(x) = a^x \ln a$

**Generally :**

If z is differentiable function with respect to $x$ , then :

* $\frac{d(e^z)}{dx} = e^z \cdot \frac{dz}{dx}$

* $\frac{d(a^z)}{dx} = a^z \ln a \cdot \frac{dz}{dx}$

**Proof:**

**1.** You have studied that the Nabier's Number (e) in form :

$$e = \frac{1}{\lfloor 0} + \frac{1}{\lfloor 1} + \frac{1}{\lfloor 2} + \frac{1}{\lfloor 3} + \cdots$$

**i.e.** $e = \sum_{n=0}^{\infty} \frac{1}{\lfloor n}$

Also, $e^x$ can be written in form : $e^x = \frac{x^0}{\lfloor 0} + \frac{x}{\lfloor 1} + \frac{x^2}{\lfloor 2} + \frac{x^3}{\lfloor 3} + \cdots$

**i.e.** $e^x = \sum_{n=0}^{\infty} \frac{x^n}{\lfloor n}$

If : $f(x) = e^x$

$$\therefore f(x) = \frac{x^0}{\lfloor 0} + \frac{x}{\lfloor 1} + \frac{x^2}{\lfloor 2} + \frac{x^3}{\lfloor 3} + \cdots$$

$$\grave{f}(x) = \text{zero} + \frac{1}{\lfloor 1} + \frac{2x}{2\lfloor 1} + \frac{3x^2}{3\lfloor 2} + \cdots$$

$$\therefore \grave{f}(x) = \frac{x^0}{\lfloor 1} + \frac{x}{\lfloor 1} + \frac{x^2}{\lfloor 2} + \cdots = e^x$$

**i.e.** $\frac{d}{dx}(e^x) = e^x$

**Generally :** If z is differentiable function with respect to $x$ , then :

$\frac{d(e^z)}{dx} = \frac{d(e^z)}{dz} \times \frac{dz}{dx}$ "chain rule" $\therefore \boxed{\frac{d(e^z)}{dx} = e^z \cdot \frac{dz}{dx}}$

**2.** (From the properties of logarithms)

$a = e^{\ln a}$

$\therefore a^x = [e^{\ln a}]^x = e^{x \ln a}$

$\therefore \frac{d}{dx}(a^x) = \frac{d}{dx}(e^{x \ln a}) = e^{x \ln a} \times \ln a$ $\therefore \boxed{\frac{d(a^x)}{dx} = a^x \ln a}$

**Generally :** If z is differentiable function with respect to $x$ , then :

$\frac{d(a^z)}{dx} = \frac{d(a^z)}{dz} \times \frac{dz}{dx}$ "chain rule" $\therefore \boxed{\frac{d(a^z)}{dx} = a^z \ln a \frac{dz}{dx}}$

## Example 1

**Find the first derivative for each of the following :**

**1.** $y = x^3 + 5e^x$

**2.** $y = x^{\frac{2}{3}} + e^{9-2x^3}$

**3.** $y = 4^x + x^4$

**4.** $y = \tan x + x^3 e^x$

**5.** $y = 2^{2x^3+5x+1}$

**6.** $y = (3x-1)e^{\tan x}$

**7.** $y = \frac{e^x}{x^2+2}$

**8.** $y = (e^{4x} - e^{-4x})^7$

**9.** $y = e^3 + \pi^{\sec x}$

**10.** $y = e^{\cos x} \times 3^{-2x}$

### Solution

**1.** $\because y = x^3 + 5e^x$ $\therefore \frac{dy}{dx} = 3x^2 + 5e^x$

**2.** $\because y = x^{\frac{2}{3}} + e^{9-2x^3}$ $\therefore \frac{dy}{dx} = \frac{2}{3}x^{\frac{-1}{3}} + e^{9-2x^3} \times (-6x^2)$

**3.** $\because y = 4^x + x^4$ $\therefore \frac{dy}{dx} = 4^x \ln 4 + 4x^3$

**4.** $\because y = \tan x + x^3 e^x$ $\therefore \frac{dy}{dx} = \sec^2 x + x^3 e^x + 3x^2 e^x$

**5.** $\because y = 2^{2x^3+5x+1}$ $\therefore \frac{dy}{dx} = 2^{2x^3+5x+1} \times (6x^2+5)\ln 2$

**6.** $\because y = (3x-1)e^{\tan x}$

$\therefore \frac{dy}{dx} = (3x-1)e^{\tan x} \times \sec^2 x + 3e^{\tan x} = e^{\tan x}[(3x-1)\sec^2 x + 3]$

**7.** $\because y = \frac{e^x}{x^2+2}$ $\therefore \frac{dy}{dx} = \frac{(x^2+2) \times e^x - e^x(2x)}{(x^2+2)^2} = \frac{e^x(x^2-2x+2)}{(x^2+2)^2}$

**8.** $\because y = (e^{4x} - e^{-4x})^7$

$\therefore \frac{dy}{dx} = 7(e^{4x} - e^{-4x})^6 \times [4e^{4x} + 4e^{-4x}] = 28(e^{4x} + e^{-4x})(e^{4x} - e^{-4x})^6$

**9.** $\because y = e^3 + \pi^{\sec x}$

$\therefore \frac{dy}{dx} = \text{zero} + \pi^{\sec x} \times \ln \pi \times \sec x \tan x$

$= \pi^{\sec x} \times \sec x \tan x \ln \pi$

**Remark**

$\because e^3$ is constant

$\therefore$ Its derivative $= 0$

**10.** $\because y = e^{\cos x} \times 3^{-2x}$

$\therefore \frac{dy}{dx} = e^{\cos x} \times 3^{-2x} \ln 3 \times (-2) + 3^{-2x} \times e^{\cos x} \times -\sin x$

$= e^{\cos x} \times 3^{-2x} [-2 \ln 3 - \sin x]$

## Derivative of logarithmic function

**1.** If $f(x) = \ln x$ , then $\grave{f}(x) = \frac{1}{x}$

**2.** If $f(x) = \log_a x$ , then $\grave{f}(x) = \frac{1}{x \ln a} = \frac{1}{x} \log_a e$

**Generally :** If z is differentiable function with respect to $x$ :

* $\frac{d(\ln z)}{dx} = \frac{1}{z} \cdot \frac{dz}{dx}$

* $\frac{d(\log_a z)}{dx} = \frac{1}{z \ln a} \cdot \frac{dz}{dx} = \frac{1}{z} \log_a e \cdot \frac{dz}{dx}$

**Proof:**

**1.** $\because y = \ln x$ $\therefore x = e^y$ (by differentiating both sides with respect to $x$)

$\therefore 1 = e^y \frac{dy}{dx}$ $\therefore \frac{dy}{dx} = \frac{1}{e^y} = \frac{1}{x}$

**i.e.** $\frac{d}{dx}(\ln x) = \frac{1}{x}$

**Generally :** If z is differentiable function with respect to $x$ , then :

$\frac{d(\ln z)}{dx} = \frac{d(\ln z)}{dz} \times \frac{dz}{dx}$ (Chain rule) $\therefore \frac{d(\ln z)}{dx} = \frac{1}{z} \frac{dz}{dx}$

**2.** $\frac{d}{dx}(\log_a x) = \frac{d}{dx}\left(\frac{\ln x}{\ln a}\right) = \frac{1}{\ln a} \cdot \frac{d \ln x}{dx} = \frac{1}{x} \cdot \frac{1}{\ln a} = \frac{1}{x} \log_a e$

because $\ll \frac{1}{\ln a} = \log_a e \gg$

**Generally :**

If z is a differentiable function with respect to $x$ ,

then : $\frac{d}{dx}(\log_a z) = \frac{d}{dz}(\log_a z) \cdot \frac{dz}{dx}$ $\quad\therefore\quad \boxed{\frac{d}{dx}(\log_a z) = \frac{1}{z \ln a} \times \frac{dz}{dx}}$

## Example 2

**Find the first derivative for each of the following :**

**1.** $y = 4x + 9 \ln x$

**2.** $y = \log_4 x$

**3.** $y = \ln (3x^4 + 8)$

**4.** $y = \log (4x^2 - 1)^3$

**5.** $y = x^4 \ln x^3$

**6.** $y = 2x + e^x \ln x$

**7.** $y = \ln \frac{x^3}{x+4}$

**8.** $y = \frac{1 + \ln x}{1 - \ln x}$

## Solution

**1.** $\because y = 4x + 9 \ln x$ $\quad\therefore \frac{dy}{dx} = 4 + 9 \times \frac{1}{x} = 4 + \frac{9}{x}$

**2.** $\because y = \log_4 x$ $\quad\therefore \frac{dy}{dx} = \frac{1}{x \ln 4} = \frac{1}{x} \log_4 e$

**3.** $\because y = \ln (3x^4 + 8)$ $\quad\therefore \frac{dy}{dx} = \frac{1}{3x^4 + 8} \times 12x^3 = \frac{12x^3}{3x^4 + 8}$

**4.** $\because y = \log (4x^2 - 1)^3 = 3 \log (4x^2 - 1)$

$\therefore \frac{dy}{dx} = 3 \times \frac{8x}{(4x^2 - 1) \ln 10} = \frac{24x}{(4x^2 - 1) \ln 10}$

**5.** $\because y = x^4 \ln x^3$

$\therefore \frac{dy}{dx} = x^4 \times \frac{1}{x^3} \times 3x^2 + \ln x^3 \times 4x^3 = 3x^3 + 4x^3 \ln x^3 = x^3 (3 + 4 \ln x^3)$

**6.** $\because y = 2x + e^x \ln x$

$\therefore \frac{dy}{dx} = 2 + e^x \times \frac{1}{x} + \ln x \times e^x = 2 + e^x \left(\frac{1}{x} + \ln x\right)$

**7.** $\because y = \ln \frac{x^3}{x+4}$

$\therefore \frac{dy}{dx} = \frac{1}{\left(\frac{x^3}{x+4}\right)} \times \frac{(x+4) \times 3x^2 - x^3 (1)}{(x+4)^2} = \frac{3x^3 + 12x^2 - x^3}{x^3 (x+4)} = \frac{2x^3 + 12x^2}{x^3 (x+4)}$

$= \frac{2x^2 (x+6)}{x^3 (x+4)} = \frac{2 (x+6)}{x (x+4)}$

**Another Solution**

$\because y = \ln \dfrac{x^3}{x+4}$

$= \ln x^3 - \ln (x+4)$ (by using logarithms rules)

$= 3 \ln x - \ln (x+4)$

$\therefore \dfrac{dy}{dx} = \dfrac{3}{x} - \dfrac{1}{x+4} = \dfrac{3x+12-x}{x(x+4)} = \dfrac{2x+12}{x(x+4)} = \dfrac{2(x+6)}{x(x+4)}$

**8.** $\because y = \dfrac{1+\ln x}{1-\ln x}$

$\therefore \dfrac{dy}{dx} = \dfrac{(1-\ln x) \times \dfrac{1}{x} - (1+\ln x) \times \dfrac{-1}{x}}{(1-\ln x)^2} = \dfrac{2}{x(1-\ln x)^2}$

## Remarks

**1.** Slope of the tangent to the curve $y = e^x$ at any point on it is the value of y-coordinate of this point.

**2.** Slope of the tangent to the curve $y = \ln x$ at any point on it is the reciprocal of $x$-coordinate of this point.

**3.** $\because \ln |x| = \begin{cases} \ln x & x > 0 \\ \ln (-x) & x < 0 \end{cases}$

⊙ at $x > 0$ , then $\dfrac{d}{dx}(\ln x) = \dfrac{1}{x}$ ⊙ at $x < 0$ , then $\dfrac{d}{dx}\big(\ln (-x)\big) = \dfrac{1}{-x} \times -1 = \dfrac{1}{x}$

**i.e.** $\dfrac{d}{dx}(\ln |x|) = \dfrac{1}{x} \; \forall \; x \neq 0$

**Generally :** If z is a differentiable function with respect to $x$

$\therefore \dfrac{d}{dx}[\ln |z|] = \dfrac{d}{dz}[\ln |z|] \times \dfrac{dz}{dx}$ (chain rule) $= \dfrac{1}{z} \times \dot{z} = \dfrac{\dot{z}}{z}$

## Example 3

**Find the first derivative for each of the following :**

**1.** $y = \ln |x^2 - 1|$ **2.** $y = \log |\sin x|$

**Solution**

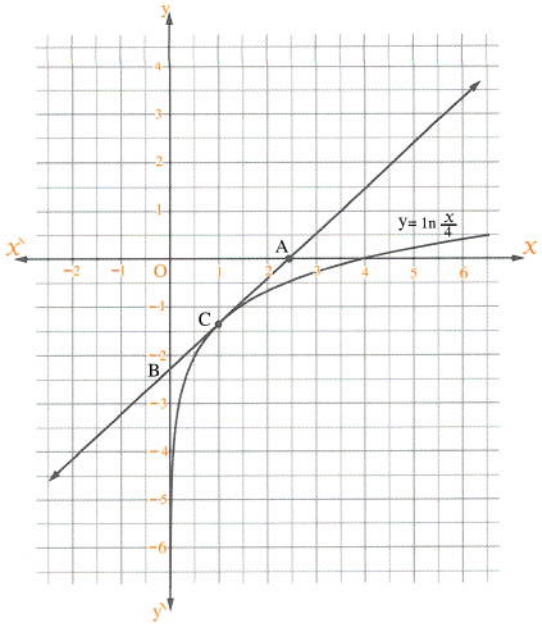
**1.** $\because y = \ln |x^2 - 1|$ $\therefore \dfrac{dy}{dx} = \dfrac{1}{x^2-1} \times 2x = \dfrac{2x}{x^2-1}$

**2.** $\because y = \log |\sin x|$ $\therefore \dfrac{dy}{dx} = \dfrac{1}{(\sin x) \ln 10} \times \cos x = \dfrac{\cot x}{\ln 10} = \cot x \log_a e$

## Example 4

**If $\overleftrightarrow{AB}$ is the tangent to the curve : $y = \ln \frac{x}{4}$ at the point $(1 , y)$ intersects $x$-axis at the point A , y-axis at the point B**

**Find length of $\overline{AB}$**




### Solution

$\because \frac{dy}{dx} = \frac{4}{x} \times \frac{1}{4} = \frac{1}{x}$

$\therefore$ Slope of the tangent to the curve at the point

$C(1 , y) = \left(\frac{dy}{dx}\right)_{x=1} = 1$

$\because \overleftrightarrow{AB}$ touchs the curve at the point $C(1 , y)$

$\therefore y = \ln \frac{1}{4} = -\ln 4$ $\quad\quad$ $\therefore C = (1 , -\ln 4)$

$\therefore$ Equation of the tangent $\overleftrightarrow{AB}$ is $\frac{y + \ln 4}{x - 1} = 1$ $\quad\quad$ $\therefore y + \ln 4 = x - 1$

to find the intersection point of $\overleftrightarrow{AB}$ with $x$-axis , we put $y = 0$

$\therefore x = 1 + \ln 4$ $\quad\quad$ $\therefore A = (1 + \ln 4 , 0)$

$\therefore$ To find the intersection point of $\overleftrightarrow{AB}$ with y-axis , we put $x = 0$

$\therefore y = -1 - \ln 4$ $\quad\quad$ $\therefore B = (0 , -1 - \ln 4)$

$\therefore AB = \sqrt{(1 + \ln 4)^2 + (1 + \ln 4)^2} = \sqrt{2}\,(1 + \ln 4)$

## Example 5

**Find the equation of the tangent and the normal to the curve : $y = e^{2x} \cos x$ at $x = 0$**

### Solution

at $x = 0$ $\quad\quad$ $\therefore y = e^0 \cos 0 = 1$

$\therefore$ The point $(0 , 1)$

$\because \frac{dy}{dx} = e^{2x} \times -\sin x + \cos x \times e^{2x} \times 2$ $\quad\quad$ $\therefore \left(\frac{dy}{dx}\right)_{x=0} = 1 \times \text{zero} + 1 \times 1 \times 2 = 2$

$\therefore$ Slope of the tangent $= 2$

$\therefore$ Equation of the tangent is $\frac{y - 1}{x - 0} = 2$ $\quad\quad$ $\therefore y - 2x = 1$

, slope of the normal $= -\frac{1}{2}$

$\therefore$ Equation of the normal is $\frac{y - 1}{x - 0} = \frac{-1}{2}$ $\quad\quad$ $\therefore 2y + x = 2$

## Example 6

If $y = e^{x} \sin x$ , prove that : $\frac{d^2 y}{d x^2} - 2\frac{d y}{d x} + 2y = 0$

### Solution

$\because y = e^{x} \sin x$

$\therefore \frac{d y}{d x} = e^{x} \cos x + \sin x \times e^{x} = e^{x} (\sin x + \cos x)$

$\therefore \frac{d^2 y}{d x^2} = e^{x} (\cos x - \sin x) + (\sin x + \cos x) \times e^{x} = e^{x} (2 \cos x)$

$\therefore \frac{d^2 y}{d x^2} - 2\frac{d y}{d x} + 2y$

$= e^{x} \times 2 \cos x - 2 e^{x} (\sin x + \cos x) + 2 e^{x} \sin x$

$= 2 \cos x \times e^{x} - 2 \sin x \times e^{x} - 2 \cos x \times e^{x} + 2 \sin x \times e^{x} =$ zero

## Example 7

If $x = \ln n$ , $y = \sin n$ , find : $\frac{d y}{d x}$ , $\frac{d^2 y}{d x^2}$

### Solution

$\because x = \ln n$ , $\therefore \frac{d x}{d n} = \frac{1}{n}$

, $\because y = \sin n$ $\therefore \frac{d y}{d n} = \cos n$

$\therefore \frac{d y}{d x} = \frac{\frac{d y}{d n}}{\frac{d x}{d n}} = \frac{\cos n}{\frac{1}{n}} = n \cos n$

, $\frac{d^2 y}{d x^2} = \frac{d}{d x}\left(\frac{d y}{d x}\right) = \frac{d}{d n} (n \cos n) \times \frac{d n}{d x} = (- n \sin n + \cos n)\, n = - n^2 \sin n + n \cos n$

### Logarithmic differentiation

The relation among the variables (in case of existence of a function as a power of another function) can be expressed in a logarithmic form by taking the natural logarithm to its both sides and using the properties of logarithms to simplify the relation before calculating the differentiation operations and the following example shows that.

## Example 8

**Find $\frac{dy}{dx}$ for each of the following :**

**1.** $y = x^x$

**2.** $y = (\sec x)^x$

**3.** $y = x^{\sin x}$

**4.** $x^y = e^{y-x}$

### Solution

**1.** $\because y = x^x$ (by taking the natural logarithm of both sides) $\therefore \ln y = \ln x^x$

$\therefore \ln y = x \ln x$ (by differentiating both sides with respect to $x$)

$\therefore \frac{1}{y} \times \frac{dy}{dx} = x \times \frac{1}{x} + \ln x$

$\therefore \frac{1}{y} \times \frac{dy}{dx} = 1 + \ln x$ (by multiplying both sides by $(y = x^x)$)

$\therefore \frac{dy}{dx} = y\,(1 + \ln x) = x^x\,(1 + \ln x)$

**2.** $\because y = (\sec x)^x$ (by taking the natural logarithm of both sides) $\therefore \ln y = \ln (\sec x)^x$

$\therefore \ln y = x \ln (\sec x)$ (by differentiating both sides with respect to $x$)

$\therefore \frac{1}{y} \times \frac{dy}{dx} = x \times \frac{1}{\sec x} \times \sec x \tan x + \ln (\sec x)$

$\therefore \frac{1}{y} \times \frac{dy}{dx} = x \tan x + \ln (\sec x)$

(by multiplying both sides $\times$ $(y = (\sec x)^x)$)

$\therefore \frac{dy}{dx} = y\left(x \tan x + \ln (\sec x)\right) = (\sec x)^x \left(x \tan x + \ln (\sec x)\right)$

**3.** $y = x^{\sin x}$ (by taking the natural logarithm of both sides) $\therefore \ln y = \ln x^{\sin x}$

$\therefore \ln y = (\sin x) \ln x$ (by differentiating both sides with respect to $x$)

$\therefore \frac{1}{y} \times \frac{dy}{dx} = \sin x \times \frac{1}{x} + \cos x \ln x$ (by multiplying both sides by $(y = x^{\sin x})$)

$\therefore \frac{dy}{dx} = y\left(\frac{\sin x}{x} + \cos x \ln x\right) = x^{\sin x}\left(\frac{\sin x}{x} + \cos x \ln x\right)$

**4.** $x^y = e^{y-x}$ (by taking the natural logarithm of both sides) $\therefore \ln x^y = \ln e^{y-x}$

$\therefore y \ln x = (y - x) \ln e$ $\therefore y \ln x = y - x$

$\therefore x = y - y \ln x = y(1 - \ln x)$

$\therefore y = \dfrac{x}{1 - \ln x}$ (by differentiating both sides with respect to $x$)

$\therefore \dfrac{dy}{dx} = \dfrac{(1 - \ln x)(1) - x\left(-\frac{1}{x}\right)}{(1 - \ln x)^2} = \dfrac{1 - \ln x + 1}{(1 - \ln x)^2} = \dfrac{2 - \ln x}{(1 - \ln x)^2}$

## Example 9

**If $y = e^x \times \sqrt{\dfrac{1-x}{1+x}}$ where $-1 < x < 1$. Prove that : $(1 - x^2)\,\dot{y} + x^2 y = 0$**

### Solution

$y = e^x \sqrt{\dfrac{1-x}{1+x}}$ (by taking the natural logarithm of both sides)

**Remember that**

* $\ln(a \times b) = \ln a + \ln b$
* $\ln e = 1$

$\therefore \ln y = \ln\left[e^x \sqrt{\dfrac{1-x}{1+x}}\right]$

$\therefore \ln y = \ln e^x + \ln\left(\dfrac{1-x}{1+x}\right)^{\frac{1}{2}}$ $\therefore \ln y = x + \frac{1}{2} \ln\left(\dfrac{1-x}{1+x}\right)$

$\therefore \ln y = x + \frac{1}{2}[\ln(1 - x) - \ln(1 + x)]$ (by differentiating both sides with respect to $x$)

$\therefore \dfrac{1}{y} \times \dot{y} = 1 + \frac{1}{2}\left(\dfrac{-1}{1-x} - \dfrac{1}{1+x}\right)$ $\therefore \dfrac{\dot{y}}{y} = 1 + \frac{1}{2}\left(\dfrac{-1 - x - 1 + x}{(1-x)(1+x)}\right) = 1 + \frac{1}{2}\left(\dfrac{-2}{1-x^2}\right)$

$\therefore \dfrac{\dot{y}}{y} = 1 - \dfrac{1}{1-x^2}$ $\therefore \dfrac{\dot{y}}{y} = \dfrac{1 - x^2 - 1}{1 - x^2}$

$\therefore \dfrac{\dot{y}}{y} = \dfrac{-x^2}{1-x^2}$ $\therefore (1 - x^2)\,\dot{y} = -x^2 y$

$\therefore (1 - x^2)\,\dot{y} + x^2 y = 0$

## Another Solution

$\because y = e^x \sqrt{\frac{1-x}{1+x}}$ (by differentiating both sides with respect to $x$)

$$\therefore \dot{y} = e^x \times \frac{1}{2\sqrt{\frac{1-x}{1+x}}} \times \frac{(1+x)\times(-1)-(1-x)\times(1)}{(1+x)^2} + \sqrt{\frac{1-x}{1+x}} \times e^x$$

$$= e^x \times \frac{1}{2\sqrt{\frac{1-x}{1+x}}} \times \frac{-1-x-1+x}{(1+x)^2} + \sqrt{\frac{1-x}{1+x}} \times e^x$$

$$= e^x \times \frac{-1}{(1+x)^2\sqrt{\frac{1-x}{1+x}}} + \sqrt{\frac{1-x}{1+x}} \times e^x$$

$$= e^x \times \sqrt{\frac{1-x}{1+x}} \left[\frac{-1}{(1+x)(1-x)} + 1\right] = e^x \times \sqrt{\frac{1-x}{1+x}} \times \left[\frac{-1}{1-x^2} + 1\right]$$

$$\therefore \dot{y} = y \times \left[\frac{-1+1-x^2}{1-x^2}\right] \qquad \therefore \dot{y} = \frac{-y\,x^2}{(1-x^2)} \qquad \therefore (1-x^2)\,\dot{y} + x^2 y = 0$$